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Does corporate digitalization improve disclosure quality?

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Abstract

Purpose – This study investigates the potential association between corporate digitization and disclosure quality, and how this relationship is moderated by non-state ownership and institutional environment.

Design/methodology/approach – Drawing on signaling theory and factors that affect disclosure quality, the authors developed a framework to study how corporate digitization is associated with disclosure quality. The proposed framework was empirically tested using a comprehensive analysis that integrated corporate-level data on digitalization, disclosure quality, and ownership structure, with regional-level data on the institutional environment. The authors employed linear panel regression models with fixed effects.

Findings – The authors found that corporate digitization is significantly and positively associated with higher disclosure quality. This positive association is particularly pronounced for non-state-owned enterprises compared to state-owned enterprises. Additionally, an improvement in the institutional environment strengthens the positive relationship between digitization and disclosure quality.

Originality/value – This work contributes to the literature on corporate digitization by empirically investigating its impact on disclosure quality. The study also extends previous research by considering the moderating roles of ownership structure and institutional environment on the digitization-disclosure quality relationship.

Keywords Digitalization, Disclosure quality, Ownership structure, Institutional environment, Signaling theory

Paper type Research paper

1. Introduction

Digital transformation involves converting information into a digital format and leveraging new digital technologies to significantly enhance business operations (Zhai *et al.*, 2022) [1]. Over the past few decades, the emergence of personal computers and the Internet has propelled digitalization forward (Adner *et al.*, 2019), profoundly influencing an organization's value creation process and overall business strategy (Salvi *et al.*, 2021). As a result, adapting business models and prioritizing digitalization has become almost imperative for companies across all industries to maintain competitiveness and avoid obsolescence (Raimo *et al.*, 2022).

Digitalization may play a crucial role in improving the quality of information disclosure by enabling companies to effectively communicate their digital strategies, practices, and

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outcomes to stakeholders. According to Salvi *et al.* (2021), digitalization is a pivotal driver capable of enhancing the efficiency and quality of various aspects of information management. Companies with a high level of digitalization tend to disclose information about their digital practices, recognizing that the quality of digitalization information disclosure is integral to overall information disclosure quality within a company, and such disclosures can highlight the company's operational efficiency and innovation potential (Anderson *et al.*, 2006; López-Arceiz *et al.*, 2019). While existing studies have explored the relationship between digital transformation and various corporate outcomes such as innovation, performance, value creation, stock price crash risk, cost, and risk-taking (Chen and Xu, 2023; Matarazzo *et al.*, 2021; Salvi *et al.*, 2021; Tian *et al.*, 2022; Wen *et al.*, 2022; Wu *et al.*, 2022; Zhai *et al.*, 2022), literature has paid little attention to the effects of digitalization on corporate disclosure strategies.

Against this backdrop, this study aims to investigate the association between digitalization and disclosure quality. Signaling theory proves to be particularly valuable in explaining firms' decisions regarding voluntary information disclosure (Spence, 1973). While signaling theory traditionally emphasizes the symbolic value of certain corporate activities as signals sent by companies to potential investors and stakeholders, it also encompasses their functional implications. Digitalization not only conveys symbolic messages about a company's technological sophistication but also substantively improves the quality and accessibility of corporate disclosures (Anderson *et al.*, 2006; López-Arceiz *et al.*, 2019; Salvi *et al.*, 2021). Firms with strong performance tend to provide additional information to the external market, either directly or indirectly, to showcase their competitive strength and attract capital (Salvi *et al.*, 2021). By leveraging signaling theory, we anticipate that corporate digitalization enhances disclosure quality due to its significant role in reducing information asymmetry (Salimyanova *et al.*, 2019; Vural *et al.*, 2020) and its unique and valuable nature as an informational resource (Gamayuni, 2015; Salvi *et al.*, 2021).

This study further delves into examining whether the association between digitization and disclosure quality is contingent upon ownership structure and institutional environment. We posit ownership structure as a moderator based on the premise that digitization facilitates an increase in information transparency and enhances transmission efficiency, thereby mitigating prevalent information asymmetry concerns, particularly pronounced within private enterprises (Dasgupta *et al.*, 2000; Tian *et al.*, 2022). Consequently, digitalization may confer unique advantages to privately controlled firms in their efforts to enhance disclosure quality compared to state-owned enterprises. Similarly, we consider the institutional environment as a moderator due to its capacity to influence organizations' responses to and implementation of digitalization initiatives (Dasgupta *et al.*, 2000; Kanashiro and Rivera, 2019). Within robust regulatory frameworks and under strong legal protections, companies are likely incentivized to adopt digital tools and technologies more effectively, amplifying their efforts to enhance disclosure quality as they contend with heightened scrutiny and accountability. While previous studies have revealed the significant impacts of ownership structure and institutional pressures on corporate behavior, their focus has primarily been on the direct effects of these factors on information disclosure quality, while also calling for the identification of indirect mechanisms (Alnabsha *et al.*, 2018; Liu and Sun, 2010; Nagata and Nguyen, 2017; Wang *et al.*, 2014). By elucidating the moderating effects of ownership structure and institutional environment, the study surpasses the existing research and determines whether these factors indirectly shape disclosure strategies.

In light of the substantial heterogeneity observed in disclosure quality across companies (Ascioglu *et al.*, 2005; Li *et al.*, 2022; Nagata and Nguyen, 2017; Reeb and Zhao, 2013), examining the aforementioned questions also helps us identify an additional determinant of such diversity. Specifically, we can investigate the contribution of ownership structure and the institutional environment to these variations. Previous scholarly work has already

established that numerous factors, including auditor compensation (Ascioglu *et al.*, 2005), director capital (Reeb and Zhao, 2013), earnings management (Lakhal, 2015; Lobo and Zhou, 2001), ownership structure (Liu and Sun, 2010; Nagata and Nguyen, 2017), and corporate characteristics (Alnabsha *et al.*, 2018; Gao *et al.*, 2016; Landgraf and Riahi-Belkaoui, 2003), exert influence on a firm's disclosure quality. However, further exploration is warranted to uncover additional sources of diversity in disclosure strategies among firms. By scrutinizing the impact of corporate digitalization on disclosure quality and how this relationship is moderated by ownership structure and the institutional environment, we can directly address this need and make significant contributions toward a deeper comprehension of the factors influencing disclosure strategies across different companies.

Using a comprehensive panel dataset encompassing 3,252 listed companies in China from 2003 to 2019, this study provides compelling evidence that corporate digitalization significantly enhances disclosure quality. Notably, we find that non-state-owned enterprises experience a stronger positive effect of digitalization on disclosure quality compared to state-owned enterprises. Additionally, we observe that the impact of digitalization on disclosure quality is reinforced in the presence of a more favorable institutional environment. These findings contribute to scholarship in several ways. First, by identifying the degree of digitalization as a novel determinant of disclosure quality, we deepen our understanding of the factors contributing to variations in disclosure practices across firms. Second, we expand our knowledge of the implications of digitalization for corporate information disclosure, enriching our comprehension of its broader impacts. Third, we shed light on a new avenue through which non-state ownership influences corporate disclosure practices, highlighting the effective utilization of digital transformation. This advances our understanding of the role played by ownership structure in shaping corporate strategies. Finally, we provide insights into how the effects of digitalization vary with the external institutional environment, offering a deeper understanding of how legal institutions shape the business outcomes of digital transformation strategies.

The subsequent sections of this paper are organized as follows. Section 2 establishes the theoretical framework and presents the formulated hypotheses. Section 3 outlines the research methodology employed in this study. Sections 4 and 5 present the primary findings and offer robustness checks. In Section 6, we engage in a comprehensive discussion on the contributions, implications, and limitations of our research.

2. Literature review and hypotheses development

2.1 Digitalization and corporate disclosure

The disclosure of information by firms serves a critical role in mitigating information asymmetry between company managers and external stakeholders (Giacosa *et al.*, 2017). Previous research has extensively investigated various factors that influence firms' disclosure strategies. For instance, Ascioglu *et al.* (2005) found that higher audit and non-audit fees are associated with lower overall disclosure quality. Reeb and Zhao (2013), on the other hand, revealed that equity-based compensation incentivizes independent directors to improve the quality of disclosure. Additionally, research exploring the realm of earnings management has consistently reported a negative impact of earnings management practices on the quality of corporate disclosure (Lakhal, 2015; Lobo and Zhou, 2001). The ownership structure of corporations has also emerged as an influential factor affecting disclosure quality, as highlighted by Nagata and Nguyen (2017), Liu and Sun (2010), and Alnabsha *et al.* (2018). Furthermore, several published studies have shed light on how corporate characteristics, such as firm age, liquidity, listing status, industry type, corporate reputation, corporate social responsibility, financing needs, and corporate governance, intersect with disclosure quality (Alnabsha *et al.*, 2018; Gao *et al.*, 2016; Landgraf and Riahi-Belkaoui, 2003). Notably, manager/

director characteristics, including audit experience, education, and social network, have also been identified as significant determinants of disclosure quality (Li *et al.*, 2022; Reeb and Zhao, 2013). Overall, this body of research has contributed valuable insights into the multidimensional nature of factors influencing disclosure quality, encompassing both contextual and individual-level variables.

However, the existing literature review identifies a research gap, as there is a lack of empirical contributions that explore the impact of corporate digitalization on information disclosure. While no empirical evidence currently exists about how digital transformation influences disclosure quality, we draw upon signaling theory to propose that digitalization may enhance the quality of information disclosure for two primary reasons. First, digitalization serves as a pivotal driver capable of improving the efficiency and quality of various aspects of information management, including input, processing, storage, and output (Salvi *et al.*, 2021). Past studies have demonstrated that agency problems and information asymmetry can arise in firms when there is a separation between cash-flow rights and control rights, particularly in situations where ownership is highly concentrated. This often leads to conflicting interests between majority and minority shareholders (Liu and Sun, 2010). Digital technologies enable companies to collect, monitor, supervise, and disclose vast amounts of business-related data instantaneously and continuously, all at low costs. These technological advancements are beneficial in enabling shareholders and potential investors to obtain accurate and timely business information. Numerous studies have documented the ability of digitalization to improve information availability and eliminate information barriers (Salimyanova *et al.*, 2019; Vural *et al.*, 2020).

Second, the degree of digitalization embodies valuable information not typically captured by financial and non-financial disclosures due to challenges in quantifying it in monetary terms or limited attention and requirements (Gamayuni, 2015; Salvi *et al.*, 2021). Signaling theory posits that the dissemination of digitalization-related information serves as a means to mitigate information asymmetry between companies possessing insights into their best practices in digitalization and potential investors (Salvi *et al.*, 2021). Highly digitalized companies are strongly incentivized to provide a more comprehensive set of digitalization-related information to the external market. Indeed, numerous studies have substantiated the pivotal role of corporate information dissemination in mitigating information asymmetries and augmenting firm value (Anderson *et al.*, 2006; López-Arceiz *et al.*, 2019). Consequently, most highly digitalized companies are motivated to disclose information regarding their digitalization level. Disclosure quality not only refers to the extent to which information disclosed by a company accurately and transparently reflects its financial condition, performance, and other relevant aspects, but also reflects the level of information disclosed by firms (Scaltrito, 2015). Thus, the quality of digitalization information disclosure is an integral aspect contributing to the overall quality of information disclosure. The circulation of digitalization-related information could contribute to the enhancement of disclosure quality and corporate performance (Salvi *et al.*, 2021). Based on the foregoing discussion, we formulate the first research hypothesis as follows:

H1. Corporations with advanced levels of digitalization are more likely to exhibit higher disclosure quality.

2.2 Digitalization, ownership structure, and corporate disclosure

Prior research underscores the significance of ownership structure in influencing corporate disclosure quality. For instance, Liu and Sun (2010) highlighted a positive correlation between state ownership in China and corporate disclosure quality, with a more pronounced effect observed in firms where ownership is distinct from control. In an analysis of annual reports from Libyan companies, Alnabsha *et al.* (2018) demonstrated a positive link between disclosure

levels and foreign, government, and institutional ownership, whereas director ownership exhibited a negative association. (Nagata and Nguyen, 2017) arrived at comparable findings, affirming the positive impact of institutional ownership on disclosure quality in Japanese firms.

However, existing research has yet to explore the moderating impact of ownership structure. The relationship between corporate digitalization and disclosure quality may be subject to moderation by ownership structure for two primary reasons. First, digitalization helps mitigate issues stemming from highly concentrated ownership, which are particularly prevalent in non-state-owned companies (Liu and Sun, 2010). Managers in private enterprises exhibit stronger incentives for income and asset expropriation from minority shareholders compared to their counterparts in state-owned enterprises. This disparity arises from the fact that private controllers, lacking government oversight, can more readily and directly appropriate the financial gains of expropriation, unlike officials in government-run entities (Chen *et al.*, 2009).

Second, digitalization plays a crucial role in reducing information asymmetry between firms and external actors in the capital market (Tian *et al.*, 2022), and information asymmetry is typically more pronounced in private enterprises than in state-owned ones (Dasgupta *et al.*, 2000). The application of digital technology enhances information transparency and transmission efficiency, effectively mitigating moral hazards arising from information asymmetry (Chen and Xu, 2023). In essence, digitalization offers distinct advantages to privately controlled firms in alleviating agency problems, minimizing information asymmetry, and augmenting disclosure quality when compared to state-owned firms. Building upon these insights, we propose the following hypothesis:

H2. Ownership structure moderates the effect of digitalization on disclosure quality, such that the impact of digitalization is more pronounced (or weaker) for non-state-owned enterprises (state-owned enterprises).

2.3 Digitalization, institutional environment, and corporate disclosure

DiMaggio and Powell (1983) postulated that a shared legal framework could induce organizational transformations, prompting entities to adapt their strategies and business operations in accordance with government laws and regulations. Consequently, organizational strategies and structures tend to mirror rules that have been institutionalized and legitimized by the government. Building upon this rationale, we focus on regulatory pressure emanating from legal institutions, signifying organizational responses to the external influence of legal regulations. Companies operating in regions with favorable institutional environments must adhere to legal and technical requisites, encompassing mandatory information disclosure within annual reports. Furthermore, governmental oversight significantly impacts organizational conduct and information. Government pressure is typically enforced through a spectrum of rules and regulations (Wang *et al.*, 2014). The concept of regulatory pressure finds empirical support. For instance, Kanashiro and Rivera (2019) discovered that firms with a Chief Sustainability Officer (CSO) exhibit higher levels of pollution emissions; however, the presence of a CSO can enhance a firm's environmental performance in the face of stringent environmental regulations. This observation underscores that regulatory enforcement enhances oversight and accountability for corporate endeavors. Similarly, Dasgupta *et al.* (2000) demonstrated that firms subjected to regulatory inspections display superior environmental performance compared to their counterparts. Therefore, the institutional environment can influence how organizations respond to and implement digitalization initiatives. In environments with robust regulatory oversight and strong legal protections, companies may be incentivized to adopt digital tools and technologies more effectively to enhance disclosure quality, as they face greater scrutiny and accountability.

H3. The institutional environment moderates the effect of digitalization on disclosure quality, wherein the impact of digitalization is enhanced with improvements in the institutional environment.

3. Data and sample

3.1 Data source

To test our hypotheses, we constructed a unique panel dataset using data collected from two primary sources: the Marketization Index of China's Provinces: NERI Report (MICPNR) and the China Security Market and Accounting Research (CSMAR). The MICPNR provides regional-level data on the institutional environment and is published by the Social Sciences Academic Press. The CSMAR, on the other hand, is a leading research-oriented database that offers comprehensive accounting and financial information for companies listed on the Chinese stock market. We utilized the CSMAR to gather corporate-level data on digitalization, disclosure quality, and ownership structure. After excluding observations with missing values, our sample consisted of 27,632 observations (3,252 corporations) across 31 provinces during the period from 2003 to 2019.

3.2 Variables

Disclosure quality. We constructed a measure of disclosure quality for each firm using a two-step process. First, we calculated the Kim and Verrecchia measure (KV index) of disclosure quality (Kim and Verrecchia, 2001). This measure gauges the sensitivity of a firm's stock price to changes in trading volume. The rationale behind this measure is that companies with higher disclosure levels are expected to rely less on trading volume for information (Ascioglu *et al.*, 2005). The KV index acts as an indicator of disclosure extent by capturing the relationship between stock price changes and trading volume. Essentially, the KV index of disclosure quality is determined by the slope coefficient in the following regression model:

$$Ln\left|\frac{Price_{id} - Price_{i,d-1}}{Price_{i,d-1}}\right| = \beta_0 + \beta_1\left(Volume_{id} - \overline{Volume}\right) + \varepsilon_{id} \quad (1)$$

$$KV\ index_i = \beta_1 \quad (2)$$

where $Price_{id}$ and $Price_{i,d-1}$ denote the closing price per share of corporate i on day d and $d-1$. Observations are dropped when no price change (i.e. $Price_{id} = Price_{i,d-1}$) occurs to permit the computation of log absolute returns. $Volume_{id}$ is the daily trading volume in shares on day d. $\overline{Volume}$ is the average annual value of trading volume in shares. ε_{id} is the error term. The coefficient of $Volume_{id} - \overline{Volume}$, i.e. β_1, is the KV measure of disclosure quality. A lower KV index indicates a higher level of disclosure quality for the firm.

Second, we computed the difference between a firm's KV index and the maximum value, defining this as the measure of disclosure quality:

$$Disclosure\ quality_i = \widehat{KV\ index} - KV\ index_i \quad (3)$$

where $\widehat{KV\ index}$ is the maximum value of KV index for all firms. $Disclosure\ quality_i$ represents firm i's disclosure quality, and a higher value results from a shock of greater disclosure for the firm. It is a market-based measure of overall disclosure quality, which could be more advantageous than the subjective accounting or financial measures of disclosure quality, such as earnings management, discretionary accruals, and restatements, adopted in the prior literature (Ascioglu *et al.*, 2005).

Digitalization. Consistent with prior research (Tian *et al.*, 2022; Wu *et al.*, 2022), we assessed the degree of digitalization within listed companies by analyzing the frequency of keywords related to digitalization in their annual reports. The defined keyword framework encompasses four dimensions: artificial intelligence, big data technology, blockchain, and cloud computing [2]. We aggregated the frequency values across these dimensions to generate a composite score, where higher scores indicate a higher level of digitalization. We acknowledged that while these technologies indeed served as significant drivers of digital transformation across many industries, there were certainly other technologies, such as virtual/augmented reality, that also contributed to the broader landscape of digitalization.

Non-state ownership. To differentiate between various corporate ownership structures, we introduced an indicator variable that signifies whether a company is state-owned or not. A value of "1" indicates non-state-owned enterprises, while "0" represents state-owned enterprises.

Institutional environment. Consistent with previous research, we employed an institutional environment index developed by the National Economic Research Institute. Widely used in social, economic, and management studies (Shi *et al.*, 2017), this index is measured at the provincial level and varies over time. A higher value of the index indicates a more favorable institutional environment.

Control variables. To account for other factors that may influence information disclosure, we incorporated several control variables based on corporate characteristics. Consistent with previous research, we controlled for the natural logarithm of total assets (Ln(Size)), proportion of independent directors, debt-to-assets ratio, ownership concentration (including the proportion of shares held by the largest shareholder and the top 10 shareholders), dual positions (where "1" signifies the CEO and managing director roles held by the same person and "0" denotes otherwise), Tobin's Q, revenue growth rate, return on assets (ROA), and the proportion of women in senior management positions.

3.3 Econometric estimations

We conducted a linear panel regression model to investigate the relationship between digital transformation and a firm's disclosure quality using the following form:

$$Disclosure\ quality_{it} = \beta_0 + \beta_1 Degree\ of\ digitalization_{it} + \gamma X_{it} + \theta_i + \varnothing_t + \varepsilon_{it} \quad (4)$$

where $Disclosure\ quality_{it}$ is the dependent variable of corporate i's disclosure quality in year t. $Degree\ of\ digitalization_{it}$ is the independent variable of corporate i's digital transformation in year t. X_{it} is a vector of covariate controls, and γ measures their impacts. θ_i is a vector of corporate fixed effects. $\varnothing_t$ is a vector of year fixed effects. Finally, ε_{it} is the error term. $Disclosure\ quality_{it}$ and $Degree\ of\ digitalization_{it}$ are scaled by subtracting the mean value from the population mean and dividing it by the population standard deviation. Therefore, β_1 in Equation (4) can be interpreted as the difference in disclosure quality caused by a one standard deviation change in the degree of digitalization.

We further estimated the following model to examine the moderating effect of non-state ownership and institutional environment:

$$\begin{aligned} Disclosure\ quality_{it} = {} & \beta_0 + \beta_1 Degree\ of\ digitalization_{it} + \beta_2 Degree\ of\ digitalization_{it} \\ & \times Nonstate\ ownership_{it} + \beta_3 Nonstate\ ownership_{it} + \gamma X_{it} + \theta_i + \varnothing_t + \varepsilon_{it} \end{aligned} \quad (5)$$

$$\begin{aligned} Disclosure\ quality_{it} = {} & \beta_0 + \beta_1 Degree\ of\ digitalization_{it} + \beta_2 Degree\ of\ digitalization_{it} \\ & \times Institutional\ environment_{it} + \beta_3 Institutional\ environment_{it} + \gamma X_{it} \\ & + \theta_i + \varnothing_t + \varepsilon_{it} \end{aligned} \quad (6)$$

where $Nonstate\ ownership_{it}$ is an indicator variable of non-state ownership. $Institutional\ environment_{it}$ denotes the index value of the institutional environment of the province where firm i is located. It is also scaled by standardizing. Other variables are defined in the same way as those in Equation (4).

4. Main results

Table 1 presents descriptive statistics and correlation metrics. It demonstrates a positive correlation between the degree of digitalization and disclosure quality (correlation coefficient = 0.04, p-value <0.01). Moreover, the correlation table indicates that multicollinearity is not a concern.

In Table 2, we present the results from Equation (4) investigating how digital transformation impacts a firm's disclosure quality. Model 1 serves as the baseline model. Hypothesis 1 posits that firms with a higher degree of digitalization are likely to disclose higher-quality information. The coefficient of the degree of digitalization is significantly positive ($\beta_1 = 0.045$, $p < 0.01$), strongly supporting Hypothesis 1.

	Mean	SD	1	2	3	4	5	6
1. Disclosure quality	7.16	0.18	1.00					
2. Degree of digitalization	26.98	54.93	0.04	1.00				
3. Non-state ownership	0.54	0.50	−0.10	0.16	1.00			
4. Institutional environment	8.63	3.26	0.05	0.23	0.24	1.00		
5. Ln(Size)	22.00	1.29	0.31	0.06	−0.26	0.14	1.00	
6. Proportion of independent directors	0.37	0.06	0.03	0.09	0.10	0.11	0.06	1.00
7. Debt-to-assets ratio	0.46	0.21	0.16	−0.10	−0.25	−0.14	0.39	−0.02
8. Proportion of shares held by the largest shareholder	35.17	14.42	−0.07	−0.12	−0.23	−0.05	0.18	0.00
9. Proportion of shares held by the top 10 shareholders	57.29	14.26	−0.25	−0.01	0.05	0.10	0.15	0.03
10. Dual position	0.23	0.42	−0.06	0.11	0.28	0.14	−0.11	0.12
11. Tobin's Q	2.07	2.82	−0.00	0.02	0.09	0.03	−0.26	0.04
12. Revenue growth rate	0.18	0.40	−0.07	0.05	0.05	−0.04	0.03	−0.00
13. ROA	0.03	0.08	−0.11	−0.00	0.04	0.03	0.05	−0.02
14. Proportion of women in senior management	0.15	0.16	−0.04	0.06	0.17	0.11	−0.11	0.06

	7	8	9	10	11	12	13	14
7. Debt-to-assets ratio	1.00							
8. Proportion of shares held by the largest shareholder	0.04	1.00						
9. Proportion of shares held by the top 10 shareholders	−0.12	0.59	1.00					
10. Dual position	−0.13	−0.07	0.03	1.00				
11. Tobin	−0.12	−0.09	−0.08	0.03	1.00			
12. Revenue growth rate	0.02	0.02	0.09	0.01	0.00	1.00		
13. ROA	−0.31	0.11	0.17	0.02	0.04	0.19	1.00	
14. Proportion of women in senior management	−0.09	−0.06	−0.00	0.08	0.05	−0.00	0.01	1.00

Table 1.
Descriptive statistics and Pearson correlation matrix

Note(s): $N = 27{,}632$
Source(s): Authors' own work

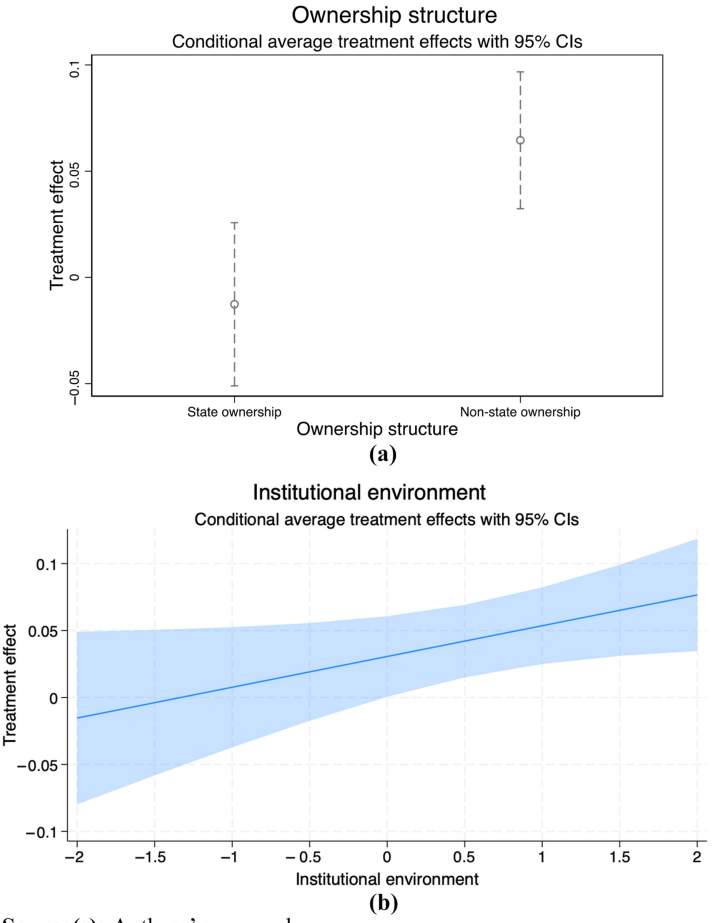
	Model 1	Model 2	Model 3
Degree of digitalization	0.045***	−0.013	0.031**
	(0.015)	(0.020)	(0.015)
Degree of digitalization * Non-state ownership		0.077***	
		(0.024)	
Non-state ownership		0.126*	
		(0.073)	
Degree of digitalization * Institutional environment			0.023**
			(0.012)
Institutional environment			−0.011
			(0.030)
Ln(Size)	0.316***	0.316***	0.315***
	(0.023)	(0.024)	(0.023)
Proportion of independent directors	−0.547***	−0.546***	−0.539***
	(0.155)	(0.155)	(0.154)
Debt-to-assets ratio	−0.268***	−0.268***	−0.271***
	(0.080)	(0.080)	(0.081)
Proportion of shares held by the largest shareholder	0.005***	0.005***	0.005***
	(0.002)	(0.002)	(0.002)
Proportion of shares held by the top 10 shareholders	−0.026***	−0.026***	−0.026***
	(0.001)	(0.001)	(0.001)
Dual position	−0.009	−0.013	−0.009
	(0.019)	(0.019)	(0.019)
Tobin's Q	0.015***	0.014***	0.014***
	(0.004)	(0.004)	(0.004)
Revenue growth rate	−0.034**	−0.036***	−0.034**
	(0.014)	(0.014)	(0.014)
ROA	−0.572***	−0.571***	−0.570***
	(0.115)	(0.115)	(0.115)
Proportion of women in senior management	0.111*	0.106	0.111*
	(0.066)	(0.065)	(0.066)
Constant	−7.348***	−7.437***	−7.377***
	(0.571)	(0.601)	(0.562)
Observations	27,632	27,632	27,632
R-squared	0.295	0.296	0.295

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses
Fixed-Effects models
Source(s): Authors' own work

Table 2.
Hypothesis testing results

Hypothesis 2 posits that the main effect is stronger for firms with non-state ownership. We evaluated Hypothesis 2 by incorporating the interaction term, *Degree of digitalization * Non-state ownership*, into Model 2. The results from Equation (5) reveal a positive coefficient for this interaction term ($\beta_2 = 0.077$; $p < 0.01$), confirming the support for Hypothesis 2. Additionally, we introduced another interaction term, *Degree of digitalization * Institutional environment*, in Model 3. The results from Equation (6) show that the coefficient of this interaction term is significantly positive ($\beta_2 = 0.023$; $p < 0.05$), indicating that the impact of digitalization on disclosure quality is heightened with a better institutional environment. Thus, this result aligns with and validates Hypothesis 3.

We present the conditional average treatment effect at different levels of the moderators, namely non-state ownership and institutional environment, based on the outcomes detailed in Models 2 and 3 of Table 2. Figure 1(a) illustrates that the impact of digitalization on disclosure quality is negative and statistically insignificant for state-owned corporations ($\beta_1 = -0.013$,



Source(s): Authors' own work

Figure 1.
The mediating effect of non-state ownership and institutional environment

$p = 0.518$), whereas this effect becomes positive and significant for non-state-owned corporations ($\beta_1 = 0.065$, $p < 0.01$). In Figure 1(b), we observe that the effect of digitalization on disclosure quality amplifies with the enhancement of the institutional environment. Similarly, the treatment effect is insignificant and negative when the institutional environment is poor (Institutional environment $= -2$), and significant and positive when the institutional environment is favorable (Institutional environment $= 2$).

5. Robustness checks

We performed a series of robustness checks to assess the robustness of the baseline results. First, we followed prior research and introduced the interactions between province dummies and year fixed effects to account for time-variant regional heterogeneities (Moser and Voena, 2012). As shown in Table 3, the estimated results remain highly consistent with the outcomes reported in Table 2.

Second, we employed the widely adopted two-stage least-squares (2SLS) regression to address concerns regarding the potential endogeneity of digitalization. The instrumental

	(1)	(2)	(3)
Degree of digitalization	0.044***	−0.013	0.032**
	(0.014)	(0.018)	(0.015)
Degree of digitalization * Non-state ownership		0.076***	
		(0.023)	
Non-state ownership		0.149**	
		(0.069)	
Degree of digitalization * Institutional environment			0.019*
			(0.011)
Constant	−7.200***	−7.352***	−7.196***
	(0.716)	(0.730)	(0.716)
Control variables	Yes	Yes	Yes
Observations	27,632	27,632	27,632
R-squared	0.326	0.327	0.326

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses. As *Institutional environment* is measured at the provincial level, it is included in the province-year fixed effect
Controlling for time-variant regional heterogeneity
Source(s): Authors' own work

Table 3.
Robustness checks

variable (IV) for corporate digitalization is a comprehensive city-level digitalization index. This index is constructed by integrating four factors related to digitalization (namely, the number of Internet broadband users per capita, the proportion of people employed in software and information technology services, the business volume of telecommunication services per capita, and the number of mobile phones per capita) using principal component analysis, providing a comprehensive reflection of a city's level of digitalization. This IV is intuitively appropriate: a corporation's level of digitalization should be highly correlated with the digitalization level of the city, while the disclosure policy of the corporation should not be directly influenced by the city's level of digitalization. The 2SLS results presented in Table 4 remain consistent with the findings reported in Table 2.

	(1)	(2)	(3)
Degree of digitalization	0.774*	0.668	0.799*
	(0.426)	(0.439)	(0.422)
Degree of digitalization * Non-state ownership		0.223***	
		(0.051)	
Non-state ownership		0.133*	
		(0.073)	
Degree of digitalization * Institutional environment			0.059**
			(0.025)
Institutional environment			−0.026
			(0.027)
Constant	−6.831***	−6.776***	−6.830***
	(0.422)	(0.425)	(0.422)
Control variables	Yes	Yes	Yes
Observations	27,632	27,632	27,632
R-squared	0.295	0.297	0.296

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses
Two-Stage least-squares regression
Source(s): Authors' own work

Table 4.
Robustness checks

Third, we employed the propensity score matching (PSM) approach to alleviate concerns about model misspecification. Our prior estimations relied on specific model specifications assuming a certain relationship between corporate digitalization and disclosure. PSM, on the other hand, does not hinge on assumptions about linear or non-linear impacts of covariates. We implemented the PSM approach through the following three steps. Initially, we categorized the sample into two groups: treatment and control groups. Firms in the treatment (control) group are those with a degree of digitalization above (below) the average. Next, we computed a propensity score for each observation based on the observed covariates using logistic regression and matched each observation in the treatment group to those in the control group using nearest neighbor matching with the closest propensity score. Lastly, we estimated Equations (3) to (5) using the sample of matched pairs. Table 5 presents the summary statistics of covariates in the treatment and control groups before and after matching. It illustrates that the mean differences of nearly all covariates decrease substantially, becoming economically negligible and statistically insignificant after matching. This finding suggests that covariates are balanced across the treatment and comparison groups after matching, indicating a high matching quality. Table 6 displays the estimation results of the matched sample. Once again, we observe that these results align with our earlier estimations.

6. Conclusion and discussion

This study offers a fresh perspective on the influence of digital technologies on organizational functions by investigating the correlation between corporate digitalization and disclosure quality. The results demonstrate a positive association between corporate digitalization and the quality of disclosure. Moreover, this relationship is notably more prominent in non-state-owned enterprises when compared to their state-owned counterparts, and it amplifies with enhancements in the institutional environment. Overall, this study identifies digitalization as a factor contributing to variability in the quality of firm disclosures and broadens the scope of digital transformation implications for business strategies.

		Treated	Control	Differences	*t*-value	*p*-value
Ln(Size)	Unmatched	22.187	21.943	0.244	13.360	0.000
	Matched	22.187	22.189	−0.002	−0.100	0.918
Proportion of independent directors	Unmatched	0.380	0.367	0.013	16.790	0.000
	Matched	0.380	0.380	0.000	0.310	0.756
Debt-to-assets ratio	Unmatched	0.412	0.468	−0.056	−19.410	0.000
	Matched	0.412	0.410	0.002	0.450	0.654
Proportion of shares held by the largest shareholder	Unmatched	32.661	35.943	−3.282	−16.140	0.000
	Matched	32.661	32.613	0.048	0.200	0.845
Proportion of shares held by the top 10 shareholders	Unmatched	58.441	56.940	1.501	7.430	0.000
	Matched	58.441	58.630	−0.189	−0.760	0.445
Dual position	Unmatched	0.331	0.203	0.128	21.560	0.000
	Matched	0.331	0.323	0.008	0.950	0.341
Tobin's Q	Unmatched	2.181	2.034	0.147	3.680	0.000
	Matched	2.181	2.134	0.047	1.070	0.287
Revenue growth rate	Unmatched	0.199	0.172	0.027	4.910	0.000
	Matched	0.199	0.197	0.002	0.330	0.741
ROA	Unmatched	0.034	0.032	0.002	2.210	0.027
	Matched	0.034	0.035	−0.001	−0.390	0.694
Proportion of women in senior management	Unmatched	0.175	0.145	0.030	13.300	0.000
	Matched	0.175	0.175	0.000	0.110	0.909

Table 5. Balancing quality of matching

Source(s): Authors' own work

	(1)	(2)	(3)
Degree of digitalization	0.056**	−0.014	0.038
	(0.022)	(0.031)	(0.025)
Degree of digitalization * Non-state ownership		0.112***	
		(0.040)	
Non-state ownership		0.009	
		(0.055)	
Degree of digitalization * Institutional environment			0.055**
			(0.023)
Institutional environment			−0.055**
			(0.026)
Constant	−6.853***	−6.814***	−6.905***
	(0.489)	(0.493)	(0.489)
Control variables	Yes	Yes	Yes
Observations	15,875	15,875	15,875
R-squared	0.288	0.288	0.288

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses
Propensity score matching method
Source(s): Authors' own work

Table 6.
Robustness checks

6.1 Contributions

This study enriches research on corporate digitalization and disclosure strategy by offering four significant contributions. First, it addresses a notable gap in our comprehension of the determinants of disclosure quality by identifying corporate digitalization as a pivotal factor. While prior research has emphasized the impact of managerial, auditor, and director characteristics on disclosure quality, showcasing diverse disclosure strategies across firms (Ascioglu *et al.*, 2005; Reeb and Zhao, 2013), there's a necessity to explore additional influential factors. By delving into the influence of corporate digitalization—an important yet underexplored factor—this study adds depth to the literature and provides empirical evidence supporting the positive correlation between digitalization and disclosure quality. Leveraging signaling theory, we contend that digitalization alleviates information asymmetry (Salimyanova *et al.*, 2019; Vural *et al.*, 2020) and represents valuable information (Gamayuni, 2015; Salvi *et al.*, 2021), consequently enhancing disclosure quality. This research contributes to understanding the determinants of corporate disclosure. This hypothesis receives empirical validation through panel structure models utilizing corporate-level data from listed companies in China. Given the substantial disparities in digitalization levels, firms with varying degrees of digital capabilities engage in information disclosure at different rates, elucidating the diversity in disclosure quality across companies. Therefore, this study significantly augments the literature on the determinants of corporate disclosure.

Second, this study advances our understanding of the ramifications of corporate digitalization by extending its effects to disclosure strategies. Previous studies have scrutinized the relationship between digital transformation and various corporate outcomes (Chen and Xu, 2023; Matarazzo *et al.*, 2021; Salvi *et al.*, 2021; Tian *et al.*, 2022; Wen *et al.*, 2022; Wu *et al.*, 2022; Zhai *et al.*, 2022). Building upon this existing research, our study demonstrates that higher degrees of digitalization are associated with enhanced disclosure quality. This finding implies that digitalization diminishes information asymmetry between firms and investors, influencing the voluntary disclosure behavior of companies. Thus, this study provides valuable insights into the repercussions of digitalization for firms.

Third, this study deepens our comprehension of the role ownership structure plays in shaping corporate strategies. While previous studies have firmly established that ownership structure significantly influences firm disclosure choices (Alnabsha *et al.*, 2018; Liu and Sun, 2010; Nagata and Nguyen, 2017; Wang *et al.*, 2014), they have primarily focused on exploring its direct effects. Our study surpasses these prior works by delving into the moderating effect of ownership structure. We find that non-state ownership amplifies the positive impact of corporate digitalization on disclosure quality. This moderation effect can be elucidated by the more pronounced agency problems and information asymmetry prevalent in non-state-owned firms compared to state-owned firms (Liu and Sun, 2010; Tian *et al.*, 2022). Consequently, digitalization can be particularly advantageous for enhancing the disclosure quality of non-state-owned companies. As a result, this study directly addresses the inquiry of why ownership structure influences the disclosure choices of firms with varying levels of digitalization.

Fourth, this study sheds light on the role of the institutional environment in shaping corporate strategies. Legal institutions wield substantial influence over organizational behavior and management (Wang *et al.*, 2014). Previous studies have demonstrated that government supervision enhances corporate disclosure and media monitoring (Dasgupta *et al.*, 2000; Xue *et al.*, 2021). In alignment with this research, our study delves into how the effect of digitalization is conditioned by the external institutional environment, deepening our understanding that legal institutions shape the business outcomes of digital transformation strategies. Our results indicate that the institutional environment moderates the effect of digitalization on disclosure quality, with a more pronounced impact observed as the institutional environment improves. Thus, this study advances institutional theory by emphasizing the significance of external legal institutions in influencing organizational behaviors.

6.2 Implications

This study holds valuable practical implications for government policymakers and corporate managers. First, it underscores that the digitalization of firms can significantly enhance their disclosure quality. Policymakers should carefully consider this positive influence of digitalization when formulating political rules and programs aimed at encouraging information disclosure. Our findings offer compelling reasoning for government involvement in subsidizing the digital transformation of companies. Additionally, corporate managers should prioritize digital transformation when devising measures to enhance company performance. Digital transformation is not just an inevitable trend in economic development but also a crucial choice for enterprises to thrive in the future. The positive impact of digitalization in reducing information asymmetry provides a solid rationale for companies to expedite their digital transformation efforts.

Second, our research sheds light on the more pronounced role of digital transformation in non-state-owned enterprises, and this influence is further magnified as the external institutional environment improves. Consequently, governments should allocate additional resources to bolster the effectiveness and fairness of legal institutions. Concurrently, companies should be mindful of the impact of ownership structure, which influences how digitalization affects disclosure quality.

6.3 Limitations

While this study significantly contributes to both theoretical and empirical research rooted in signaling theory, it is essential to acknowledge and address several limitations. First, the study's exclusive focus on listed firms in China raises the necessity for future research incorporating data from other countries to ensure the generalizability of these findings. Second, potential concerns may arise regarding the adoption of a market-based measure of disclosure quality. Although we utilized the KV index (Kim and Verrecchia, 2001) to measure disclosure quality, the

ongoing debate surrounding various measures necessitates the validation of our results using alternative metrics (Chen *et al.*, 2015). While this market-based measure offers a valuable perspective, there are potential benefits of complementing it with a more direct assessment of the content and quality of disclosures. We have incorporated an alternative measure of disclosure quality based on the annual ratings issued by the Shenzhen Stock Exchange (SZSE) and Shanghai Stock Exchange (SHSE). Due to data limitations, we are unable to incorporate a more direct measure of disclosure content in the current study. Third, it is important to note that the data on corporate digitalization initiatives is derived from self-reported information in company annual reports. While this approach provides valuable insights into firms' digital transformation efforts, we acknowledge that these self-reported measures may be subject to inherent biases and potential over- or under-reporting. Future research could benefit from incorporating more objective measures of digitalization, such as independent audits or third-party assessments, to enhance the validity and robustness of the findings. Despite these acknowledged limitations, our study lays the foundation for valuable future research opportunities. Given the increasing prevalence of digital transformation across industries, exploring its impact on corporate outcomes beyond disclosure strategies holds significant promise. Moreover, future studies focusing on customer demand may yield more conclusive findings regarding the effects of corporate digitalization, provided more appropriate data becomes available.

Notes

1. While the terms "digital transformation" and "digitalization" do not have explicit or unified definitions and are often used interchangeably in existing research (Gong and Ribiere, 2021), recent studies conclude that digital transformation encompasses more than digitalization (Gong and Ribiere, 2021; Kraus *et al.*, 2022; Verhoef *et al.*, 2021). In this study, we did not intend to conflate these terms or present them as synonymous. Instead, we aimed to explore the implications of digitalization efforts on disclosure quality, recognizing that these efforts are often part of a broader digital transformation journey undertaken by organizations.

2. The four technologies encompass a broad spectrum of capabilities crucial for digitalization. Artificial intelligence stands as a cornerstone of digitalization, providing the capacity to process vast amounts of data, discern patterns, and autonomously make intelligent decisions. Through machine learning algorithms, Artificial intelligence facilitates systems to adapt and learn from data, fueling applications like predictive analytics and natural language processing. Big data technology tackles the challenges linked with managing and analyzing extensive datasets. By utilizing tools such as Hadoop and Spark, organizations can extract valuable insights from diverse data sources, facilitating informed decision-making and fostering innovation. Blockchain technology introduces a decentralized and secure framework for digital transactions. By establishing an immutable ledger system across a distributed network, blockchain enhances transparency, security, and trust in digital interactions, thereby stimulating innovation and enabling new business models. Cloud computing revolutionizes the accessibility and utilization of computing resources for businesses. With its scalable and flexible nature, cloud computing empowers organizations to deploy applications, store data, and run workloads without the constraints of on-premises infrastructure, thus expediting digital transformation initiatives. While these technologies indeed serve as significant drivers of digital transformation across many industries, there are certainly other technologies, such as virtual/augmented reality, that also contribute to the broader landscape of digitalization. Thus, these four specific technologies do not encompass the full spectrum of technologies pertinent to digitalization.

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Appendix

We also conducted three additional robustness checks. First, we defined an alternative measure of disclosure quality based on the annual ratings issued by the Shenzhen Stock Exchange (SZSE) and Shanghai Stock Exchange (SHSE). The SZSE began assessing the quality of information disclosure for listed firms in the fiscal year of 2005, and the SHSE started in 2016. The rating for disclosure quality has four grades: Excellent (A), Good (B), Fair (C), and Poor (D), denoted as 4, 3, 2, and 1, respectively. Our results using this alternative measure remain consistent (See Table A1). Second, we additionally controlled for the proportion of foreign tradable shares and proportion of overseas income to represent external market factors, ownership structure to capture corporate governance aspects, and the institutional environment index developed by the National Economic Research Institute as a proxy for regulatory influences. The results, as reported in Table A2, show that the marginal effect of digitalization on disclosure quality remains largely consistent after incorporating these additional controls into our model. This observation is attributed to the control of firm and year fixed effects, which have already accounted for most characteristics. Third, we conducted additional robustness checks by lagging the independent variable by one, two, and three periods. The results of these robustness checks are detailed in Models 2 to 3 of Table A3. The results remain consistent across these alternative specifications.

Table A1.
Robustness check with alternative measure of disclosure quality

	Model 1	Model 2
Degree of digitalization	0.045***	0.016***
	(0.015)	(0.005)
Observations	27,632	18,062
Control variables	Yes	Yes
R-squared	0.295	0.193

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses. In Model 1, the dependent variable is the market-based measure of disclosure quality. In Model 2, the dependent variable is the rating for disclosure quality issued by SZSE and SHSE

Source(s): Authors' own work

	Model 1	Model 2
Degree of digitalization	0.045***	0.045***
	(0.015)	(0.015)
Proportion of foreign tradable shares		−0.150
		(0.170)
Proportion of overseas income		0.029
		(0.068)
Non-state ownership		0.119
		(0.073)
Institutional environment		−0.014
		(0.029)
Control variables	Yes	Yes
Observations	27,632	27,632
R-squared	0.295	0.295

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses
Source(s): Authors' own work

Table A2.
Robustness check with additional controls

	Model 1	Model 2	Model 3	Model 4
Degree of digitalization	0.045***			
	(0.015)			
L1. Degree of digitalization		0.023**		
		(0.012)		
L2. Degree of digitalization			0.022***	
			(0.006)	
L3. Degree of digitalization				0.020*
				(0.012)
Control variables	Yes	Yes	Yes	Yes
Observations	27,632	23,069	19,966	17,496
R-squared	0.295	0.260	0.214	0.159

Note(s): Level of statistical significance: ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$; Robust standard errors clustered at the corporate level are reported in parentheses
Source(s): Authors' own work

Table A3.
Robustness check with lagging effects

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